

Fractal Conscious Perception

Multiscale Morphology, Statistical Self-Similarity, and Route-Specific Neural Reinstatement

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August 13, 2026

Preprint; not peer reviewed.

Abstract

When a person recognizes a face, remembers a room, or imagines the taste and texture of broccoli, microscopic learned structure helps organize a distributed reconstruction without expanding into a brain-sized copy. The scientific problem is to explain how relations encoded at one scale constrain later patterns at other scales while the physical carriers change. Neural morphology, connectome topology, electrophysiological time series, and cortical folding each admit distinct finite-range scaling measurements; memory research independently demonstrates distributed engrams and content-sensitive reinstatement. The unresolved question is whether measured scaling properties add predictive or causal value to cross-scale reinstatement beyond ordinary morphology, topology, activity history, oscillatory variables, and flexible nonlinear alternatives. This paper develops that test before naming the proposed operation. Learning changes local cellular structure and readiness. A cue recruits only part of that structure. Receivers transform the arriving activity according to their own state, inhibition, timing, and connectivity. Recursive exchange preserves some relations, suppresses others, and re-expresses a usable pattern across changing populations. The historical Self-Aware Networks term **FRACTAL Conscious Perception-Functional Recursive Activity Cortical Telescoping Asymmetric Lensing** names this asymmetric cross-scale transformation. “Fractal” is restricted here to declared finite-range statistical relations with explicit estimators, intervals, nulls, and alternatives; it does not mean exact self-copying or one universal exponent. The framework separates morphology, topology, temporal dynamics, and functional reinstatement; formalizes added-value, relation-preservation, perturbation, and route-specific-restoration tests; and specifies four biological protocols with independent-unit, multiplicity, and safety contracts. Successive synthetic studies expose both strengths and failures. A planted feature-level generator favored the full typed model, whereas explicit graph-temporal and independent-generator studies reversed key rankings, revealed effective-capacity mismatch, and showed that an earlier recovery fraction could reward destructive collapse more than invertible transformation. Seventeen denominator-safe invariants were subsequently checked in Lean without admitted proofs or user axioms; these clarify algebra and do not establish biology. A new frozen route-identification application then evaluated four generator families, six models, five interventions, twenty independent seeds, 480 model rows, and 100 intervention rows. Two complete executions reproduced byte-for-byte. All ten prospectively declared gates passed. In the structured family, typed routes added mean information gain $\Delta_Z = 1.010752$ over preserved common context, with a 95% seed-cluster interval [0.996080, 1.026233]. Invertible rotation preserved the gain; route erasure reduced it to zero; exact matched restoration returned it; nonspecific restoration did not. The model was adverse under no-route and decoy-shift controls, and a direct single-scale model decisively won when only one scale carried the target. Raw ridge also slightly outperformed typed features in the structured family. Thus the application establishes bounded route-specific identification

and refusal behavior in a frozen synthetic system, not universal model superiority, biological implementation, conscious content, or therapeutic efficacy. The future biological evaluation remains unopened.

Keywords: fractal neuroscience; dendritic morphology; neural reinstatement; memory; finite-range scaling; route identification; NAPOT; phase-wave differential; consciousness; connectome; multifractal dynamics

1. Begin with the experience

Look at a red cup, then close your eyes and remember it. The remembered cup can retain color, shape, location, emotional significance, and the motor possibilities associated with reaching for it, even though the retinal input is gone. A similar reconstruction occurs when the word *broccoli* evokes a visual form, a texture, a smell, a taste, and learned actions. The remembered object is distributed across sensory, associative, affective, and motor systems, yet it can be experienced as one related event.

The physical storage changes supporting that event are microscopic. Synaptic efficacy, dendritic excitability, receptor composition, inhibitory routing, glial state, and cellular morphology can all change through learning. The experienced reconstruction is macroscopic and distributed. It engages populations, regions, and recurrent pathways. The unexplained step is therefore not simply storage and not simply activation. It is **cross-scale reinstatement**: how a local, learned disposition helps organize a larger pattern that preserves usable relations while changing physical form.

A magnifying glass is a tempting metaphor, but it is incomplete. Neural populations do not merely enlarge a microscopic picture. Each receiving population has different connections, delays, inhibition, receptor state, and prior activity. The larger pattern must be a transformed reconstruction. It may preserve that two features belong together, that one feature preceded another, or that a particular action follows, while using different cells, frequencies, amplitudes, and spatial extents.

The paper asks a concrete question:

Do finite-range morphological, topological, or dynamical scaling measurements predict the fidelity, transformation, or failure of cross-scale neural reinstatement better than matched nonfractal, single-scale, topology-only, oscillation-only, and history-aware models?

That question is narrower than “is the brain a fractal?” and scientifically stronger. It names an outcome, competing models, added-value criterion, and a route to causal intervention.

2. What established neuroscience already supplies

2.1 Branching morphology is measurable and computationally consequential

Dendrites are branching biological structures whose geometry controls which signals can arrive together, how strongly distal inputs influence the soma, where nonlinear events occur, and how synapses can be distributed in space. Three-dimensional analysis of rat neurons found that dendritic arbors exhibit finite-range fractal-like scaling and that an effective fractal dimension summarizes

a trade-off between connectivity and material or energetic cost (Smith et al., 2021). The scaling range was limited-approximately one order of magnitude in that preparation-which is precisely why a biological paper must report the fitted interval rather than invoke an ideal mathematical fractal.

Morphology is not merely decorative. Human layer-5 pyramidal neurons show strong dendritic compartmentalization, and the length and conductance distribution of the arbor alter input-output behavior (Beaulieu-Laroche et al., 2018). Recent modeling and electrophysiological work further connects human dendritic morphology and synaptic nonlinearities to functional complexity (Aizenbud et al., 2026). These results make a morphology-to-computation bridge plausible, while leaving open which geometric statistic best predicts a particular readout.

2.2 Circuit topology has its own scaling description

At the network level, Bassett and colleagues found physical and topological Rentian scaling in *C. elegans*, human cortical anatomical networks, and engineered circuits (Bassett et al., 2010). Rentian scaling measures how the number of connections crossing a partition boundary changes with the number of nodes inside the partition. This is not the same object as dendritic box-counting dimension. A circuit can have a scale-dependent topology without every cell sharing one morphological exponent.

The distinction matters because wiring geometry, graph topology, and neural dynamics can covary. A useful test must determine whether a scaling measurement adds information after simpler variables-degree, path length, modularity, distance, density, and cell type-are included.

2.3 Neural time series exhibit multiple forms of scale dependence

Amplitude envelopes of human 10 and 20 Hz oscillations can show long-range temporal correlations over many cycles (Linkenkaer-Hansen et al., 2001). Detrended fluctuation analysis provides one way to estimate this form of temporal dependence, but its result depends on preprocessing, stationarity, window selection, and the chosen scale interval (Hardstone et al., 2012). Intracranial EEG work comparing monofractal and multifractal measures showed that uncorrected metrics can be strongly correlated with signal variance; after epoch-wise standardization, multifractal measures could retain nonredundant information (França et al., 2018).

A power-law-looking plot is therefore a hypothesis, not a conclusion. Maximum-likelihood estimation, goodness-of-fit tests, alternative distributions, and an explicit lower cutoff are needed (Clauset, Shalizi, and Newman, 2009). Power-law avalanche statistics can also arise away from criticality and even from independent stochastic surrogates (Touboul and Destexhe, 2017). The present framework treats geometric fractality, temporal long-range dependence, multifractality, avalanche scaling, aperiodic spectral slope, and criticality as related but nonidentical measurements.

2.4 Memory is distributed and retrieval reinstates patterns

Brain-wide mapping in mice identified a distributed complex of engram-cell ensembles for one contextual fear memory and found that simultaneous activation of multiple component ensembles produced stronger recall than activation of a single ensemble (Roy et al., 2022). Human intracranial recordings show trial-specific reinstatement of distributed oscillatory power during successful retrieval, with retrieval sometimes unfolding faster than encoding (Yaffe et al., 2014). Human fMRI studies likewise report item-specific encoding-retrieval similarity for individual scenes and coordinated

reinstatement between hippocampal and cortical regions (St-Laurent et al., 2015; Bosch et al., 2014; Jonker et al., 2018).

These findings establish distributed reinstatement as a measurable outcome. They do not require the recalled pattern to be identical to the encoding pattern. A reconstruction may be compressed, accelerated, completed from a partial cue, or expressed in a different population state while preserving behaviorally relevant relations.

2.5 Scaling measurements can covary with conscious state

Fractal dimension, spectral slope, avalanche measures, Lempel-Ziv complexity, and perturbational complexity can change across wakefulness and anesthesia. In one controlled anesthesia dataset, EEG fractal dimension decreased under propofol and xenon but not under ketamine, where participants could remain experientially conscious through vivid dreams despite behavioral unresponsiveness (Maschke et al., 2024). Near-critical slow cortical dynamics have also been associated with conscious state across anesthesia, psychedelic, sleep, and disorders-of-consciousness datasets (Toker et al., 2022).

These studies make state-dependent scaling an important measurement domain. The present paper adds a content-sensitive question: can typed scaling variables predict *which distributed pattern is reinstated*, how it is transformed, and whether it reaches report, recognition, or action? State and content must be modeled separately.

3. Source genealogy inside Self-Aware Networks

The current formalization is new, but its problem and mechanism developed through a dated SAN sequence. Each stage should receive only the content it actually contains.

3.1 Early network and morphology ancestors, 2012

In August 2012, Blumberg publicly described continuous scales of communication from local neural groups and hemispheres to interactions between people. The associated dialogue used “fractal symmetry,” but that exact wording came from Ben Thomas; Blumberg’s attributable contribution is the proposed continuity of network communication across local and larger scales. In September 2012, *Neurons are Programs Transmitting Their Shape Theory* proposed that a neuron’s learned topology functions as a program, that synaptic history contributes to that topology, and that the emitted analog event carries a compressed consequence of the cell’s learned structure and temporal state. These are early ancestors of morphology-conditioned receive-transform-project reasoning. They do not contain the later FRACTAL acronym, NAPOT, or the completed cross-scale estimator.

3.2 Public sensorimotor and recursive-feedback bridge, 2017-2018

The April 2017 *Neural Lace Podcast* framed neural communication as a mixture of connection-and-feedback traffic and broadcast-like traffic, named **Alternating Transmission Protocol**, and paired the acronym ATP with adenosine triphosphate. The same 2017 program described a time-varying

three-dimensional ionic or electromagnetic matrix and a spatial computation that combines features into predictive object representations.

The September 2017 article *Getting 3D Shapes for VR from 2D Images with AI to Make a Self-Aware Robot's Brain* proposed building an internal three-dimensional space from two-dimensional inputs, linking words and sounds to spatial concepts, and transferring broad, potentially lossy representations among neural bundles. In 2018, *Addressing Criticism for "Humans Are Metal Robots"* described perception as a rendered concept distributed through a **fractal of feedback loops**, joined coincidence detection and subthreshold dendritic computation, and compared the target architecture to a neural network combined with a three-dimensional point cloud and recursive feedback.

This stage establishes recursive, embodied, spatial, and multiscale traffic as a public SAN bridge. It precedes the later named tomography and phase-wave operators.

3.3 The Watercolor Neuron Signals recording, 2021

The Google Recorder source *Watercolor Neuron Signals* is dated October 24, 2021 by embedded M4A creation metadata and preserves the original public Recorder route. It states the cross-scale problem unusually directly. Multiple dendrites capture partial patterns; firing overlays them “like watercolor”; coincident inputs contribute different pieces; neurons that fire together begin to oscillate together and interact with a larger oscillating group; and a network can track coincident patterns across scales. The recording also proposes tomography from synaptically captured patterns and compares the combination of partial neural perspectives with projection-slice reconstruction.

The recording is a machine-transcribed source whose load-bearing wording requires audio readback before quotation. Its service and media metadata support the 2021 recording date; the public Git transcript fixation begins June 8, 2022. It supplies a strong conceptual ancestor of the present operator without backdating the later acronym or equations.

3.4 Mature 2022 SAN stages

Public Git history fixes a denser mechanism during June-August 2022:

- **a0061z** describes compressed feature representations, dendritic and network-scale pattern selection, and phasic activity in multilevel fractal patterns.
- **a0142z** distinguishes coherence used for coupling from phase separation used for distinction, joins morphology and inhibition to memory, and proposes scale-invariant tempo-spatial patterns.
- **a0337z** asks whether a locally stored pattern gains dimensionality when activated and transmitted to an oscillating array, with excitatory and inhibitory phase changes contributing to a layered perceptual field.
- the NAPOT page states that memory scales from synaptic connections to whole-brain oscillatory information flow and treats arrays as successive receivers and projectors.
- the Fractals page explicitly joins morphology, medical imaging, scale-dependent activity, and the movement from a memory stored in a synapse to whole-brain focus.

The SAN Whitepaper names **FRACTAL Conscious Perception: Functional Recursive Activity Cortical Telescoping Asymmetric Lensing** in a version immutably Git-fixed on

August 24, 2022. It proposes that small, learned dispositions become differentiated circuit- and brain-scale activity; the larger patterns are asymmetric transformations rather than perfect mirrors. Four public 2022 video explanations join synapse-to-network scaling, selective reception, array-to-array projection, oscillatory coordination, volumetric reconstruction, and distributed observation.

3.5 Later sharpening, 2024-2026

The 2024 *Analog Biological Backpropagation* article adds an action-dependent learning bridge: outcomes can alter which local structures and pathways become easier to recruit later. The 2026 *Telescoping Network Reinstatement* paper formalizes a local-to-population reconstruction operator and supplies finite synthetic reconstructions, while *From Morphology to Phase* formalizes morphology-dependent receiver state. Neither makes fractal measurement and its added predictive value the main endpoint.

The present paper therefore does not merely rename those works. It asks whether measured finite-range scaling properties explain variance in reinstatement that those earlier models leave unassigned.

4. The operation before the name

The mechanism can be stated without specialized vocabulary:

1. Learning changes synapses, dendritic branches, inhibitory routes, receptor state, and network connectivity.
2. A later cue activates only part of that learned structure.
3. The activated cell does not transmit its morphology. Its morphology conditions an output event.
4. Multiple output events arrive at another population with different timing, strength, and pathway context.
5. The receiving population combines them according to its own learned structure and current state.
6. Recurrent exchange stabilizes some relations and suppresses others.
7. The process repeats across successively larger or differently organized populations.
8. The final distributed pattern can preserve object identity, sequence, orientation, value, or action relevance without physically copying the first local state.
9. Behavior or report tests the reconstruction, and returned sensory consequences modify future processing.

The key word is **relation**. A triangle can be represented by pixels, edge detectors, population activity, spoken language, or motor trajectories. The carriers differ, but some relations can survive the transformation. Cross-scale neural reinstatement should therefore be tested by relation preservation and predictive consequence, not by demanding cell-for-cell identity.

5. Earning the historical term

The preceding operation gives a precise role to the historical phrase **FRACTAL Conscious Perception**.

- **Functional:** the proposed similarity must improve prediction or causal control of a neural or behavioral outcome.
- **Recursive Activity:** receiving populations transform and return activity through recurrent pathways.
- **Cortical Telescoping:** a local learned disposition influences a wider distributed reconstruction through successive transforms.
- **Asymmetric Lensing:** each scale emphasizes, compresses, suppresses, or completes different relations; the result is not an identical copy.

“Fractal” is used only when a declared measurement shows finite-range self-similarity, self-affinity, multifractality, Rentian scaling, or another explicitly defined scaling relation against appropriate nulls. Functional recurrence without such a measurement remains recursive multiscale processing, not automatically fractal processing.

The resulting candidate mechanism is:

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learned local morphology and state
-> cue-dependent local readout
-> receiver-relative phase and amplitude differences
-> scale-indexed asymmetric transformations
-> recurrent selection and relation preservation
-> distributed reinstatement
-> conscious recognition, report, or action
-> returned evidence and further learning
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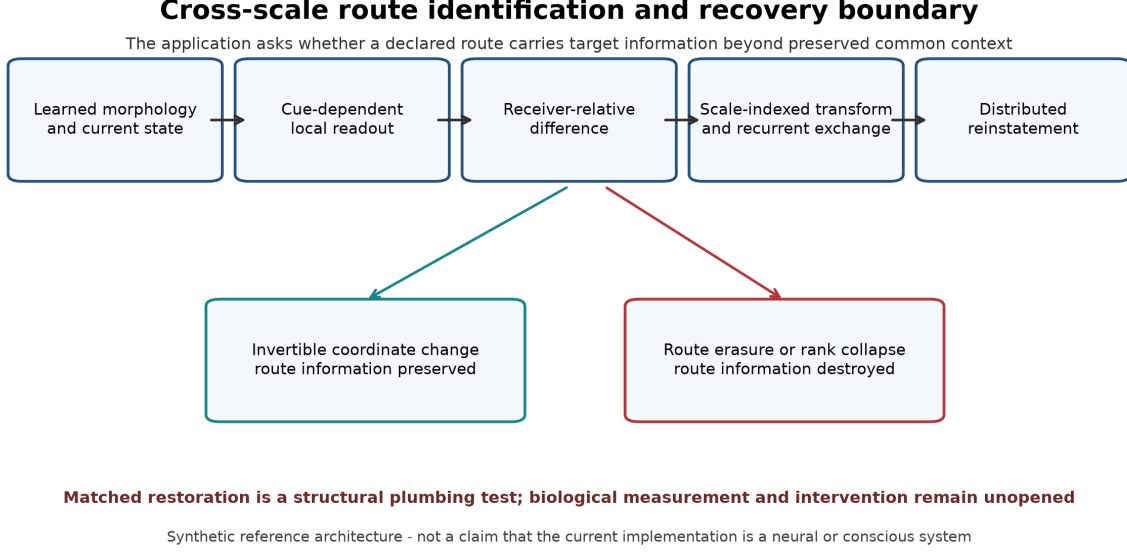


Figure 1: Cross-scale route architecture. Learned morphology and current state condition a local readout; receiver-relative differences are transformed through recurrent scale-indexed routes into distributed reinstatement. The lower branch distinguishes invertible coordinate change from information-destroying erasure.

6. Typed mathematical framework

6.1 Measurement objects and finite scale intervals

Let $s \in \mathcal{S}$ denote a declared neural scale or representation level, not an assumption that the levels share one geometry. At each s , define a measured object

$$X_s = (M_s, G_s, A_s, Y_s), \quad (1)$$

where M_s is morphology, G_s is graph topology, $A_s(t)$ is activity, and Y_s is a functional endpoint. Every estimator is paired with a finite admissible interval $I_s = [\epsilon_{\min,s}, \epsilon_{\max,s}]$ or time range $[n_{\min,s}, n_{\max,s}]$.

6.2 Morphological box-counting and local slopes

For a segmented arbor or tissue surface, let $N_s(\epsilon)$ be the number of boxes of side ϵ intersecting the object. A candidate finite-range scaling law is

$$N_s(\epsilon) = C_s \epsilon^{-D_s} \exp\{r_s(\epsilon)\}, \quad \epsilon \in I_s, \quad (2)$$

where r_s is a residual. Rather than trust one fitted line, estimate the scale-dependent slope

$$D_s(\epsilon) = -\frac{d \log N_s(\epsilon)}{d \log \epsilon}. \quad (3)$$

A stable plateau in $D_s(\epsilon)$, adequate range, bootstrap confidence intervals, and superiority to alternative geometric models are required before a fractal description is accepted.

6.3 Lacunarity and branching structure

Two arbors can share a box-counting dimension while distributing gaps differently. If $m_{s,k}(\epsilon)$ is occupied mass in box k , define lacunarity

$$\Lambda_s(\epsilon) = \frac{\text{Var}_k[m_{s,k}(\epsilon)]}{\text{E}_k[m_{s,k}(\epsilon)]^2} + 1. \quad (4)$$

The morphological feature vector should therefore include D_s , Λ_s , Sholl curves, branch order, path length, fork angles, taper, synaptic density, and cell type. The scaling statistic must compete with, rather than stand in for, these ordinary features.

6.4 Temporal long-range dependence and multifractality

For an activity-derived signal $a_s(t)$, detrended fluctuation analysis estimates

$$F_s(n) \propto n^{H_s}, \quad n \in [n_{\min,s}, n_{\max,s}], \quad (5)$$

where H_s indexes long-range dependence under the declared preprocessing. For multifractal analysis, a partition function may be written

$$Z_s(q, \epsilon) = \sum_k \mu_{s,k}(\epsilon)^q \propto \epsilon^{\tau_s(q)}. \quad (6)$$

Linearity of $\tau_s(q)$ suggests a monofractal description; curvature motivates a multifractal model. Phase-randomized, amplitude-adjusted, end-matched, and block-shuffled surrogates test whether apparent multifractality follows from the spectrum, marginal distribution, cyclic boundaries, or nonstationarity.

6.5 Network scaling

For a graph partition containing (n) nodes with (e_{∂}) edges crossing its boundary, topological Rentian scaling is

$$e_{\partial} = Kn^p + \xi, \quad (7)$$

over a fitted partition range. The exponent p describes a network property and is not equated with dendritic D_s or temporal H_s .

6.6 Receiver-relative local differences

Let neuron or population i receive an event with phase $\phi_i(t)$. Its learned state predicts a route-specific relation $\hat{\phi}_i(t \mid h_i, m_i, g_i)$, conditioned on activity history h_i , morphology m_i , and graph context g_i . Define the circular residual

$$\delta_i(t) = \text{wrap} \left[\phi_i(t) - \hat{\phi}_i(t \mid h_i, m_i, g_i) \right]. \quad (8)$$

This receiver-relative departure is one measurable component of a Phase-Wave Differential. Its biological consequence must be modeled separately through changes in excitability, timing, downstream recruitment, or plasticity.

6.7 Scale-indexed transformation

Let $z_s(t)$ be a latent relational state estimated at scale s . A cross-scale step is

$$z_{s+1}(t + \Delta_s) = T_s(z_s(t), \delta_s(t), m_s, g_s, h_s) + \eta_s(t), \quad (9)$$

where T_s may be nonlinear and lossy. The sequence $T_{S-1} \circ \dots \circ T_1$ is the telescoping operator. ‘‘Asymmetric lensing’’ means that T_s is not assumed to be an isometry or literal magnifier.

6.8 Relation preservation

Let R_s be a matrix of task-relevant pairwise relations among features represented at scale s , and let R_{s+1} be the corresponding relation matrix after transformation. Define normalized relation preservation

$$\rho_s = 1 - \frac{\|W_s \odot (R_{s+1} - \Pi_s R_s \Pi_s^\top)\|_F}{\|W_s \odot \Pi_s R_s \Pi_s^\top\|_F + \varepsilon}, \quad (10)$$

where Π_s maps feature identities or correspondences across levels and W_s weights relations relevant to the task. This permits carrier changes while demanding measurable preservation of selected structure.

6.9 Reinstatement fidelity

For encoding pattern e_j and retrieval pattern r_j on trial j , define a cross-validated reinstatement score

$$\mathcal{R}_j = \text{sim}(Q_{\text{enc}} e_j, Q_{\text{ret}} r_j) - \mathbb{E}_{k \neq j} \text{sim}(Q_{\text{enc}} e_j, Q_{\text{ret}} r_k), \quad (11)$$

where the mappings $Q_{\text{enc}}, Q_{\text{ret}}$ are fitted only on training data. The subtraction makes the endpoint item-specific rather than a generic state similarity.

6.10 Added predictive value

Let B_j contain baseline variables: cell type, total length, branch count, synaptic density, graph degree, path length, firing rate, oscillatory power, activity history, and task condition. Let F_j contain the typed scaling features. Compare

$$\mathcal{M}_0 : \mathcal{R}_j = f(B_j) + \epsilon_j, \quad \mathcal{M}_1 : \mathcal{R}_j = f(B_j, F_j) + \epsilon_j. \quad (12)$$

The primary added-value endpoint is nested, held-out risk reduction

$$\Delta_{\text{scale}} = \text{Risk}_{\text{test}}(\mathcal{M}_0) - \text{Risk}_{\text{test}}(\mathcal{M}_1). \quad (13)$$

The paper’s central claim earns support only when $\Delta_{\text{scale}} > 0$ with a preregistered confidence interval and survives matched surrogates and ablations.

6.11 Conscious access and content

Let C_j denote a content-sensitive outcome such as correct recognition, forced-choice identity, confidence under controlled response bias, or a no-report perceptual marker. The scale features add content information only if

$$I(C; F \mid B, \mathcal{R}_{\text{generic}}) > 0 \quad (14)$$

on held-out data, where $\mathcal{R}_{\text{generic}}$ captures nonspecific arousal or global-state similarity. This separates content prediction from merely distinguishing wakefulness from anesthesia.

6.12 Perturbation and rescue

For a targeted intervention P , matched sham S , and rescue U , define

$$\Delta_P = \mathcal{R}_S - \mathcal{R}_P, \quad \Delta_U = \mathcal{R}_U - \mathcal{R}_P. \quad (15)$$

The causal pattern predicted by the model is $\Delta_P > 0$ and $\Delta_U > 0$, with rescue selective for the relations and pathway affected rather than a global increase in excitability.

6.13 Multiscale sufficiency score

A single headline index is permitted only after every component is reported separately. One bounded summary is

$$\mathcal{F}_{\text{FCP}} = \Delta_{\text{scale}} + \lambda_1 \bar{\rho} + \lambda_2 I(C; F \mid B, \mathcal{R}_{\text{generic}}) + \lambda_3 \Delta_U - \lambda_4 \Omega, \quad (16)$$

where Ω penalizes model complexity, unstable scale ranges, preprocessing sensitivity, and surrogate failures. Before aggregation, every positively oriented term is normalized on a preregistered training distribution and every $\lambda_k \geq 0$ is fixed without access to final-test outcomes. The index is a comparison tool, not a consciousness meter.

6.14 Estimator and source-boundary contract

The estimator contract treats each scale statistic as an estimator with a declared object, preprocessing path, admissible range, uncertainty method, and null model. A fitted exponent is not a mechanism, and a visually straight log-log plot is not sufficient evidence of self-similarity. The primary estimator record must include:

1. the measured object and segmentation or signal-extraction method;
2. the finite scale or time interval over which the statistic is evaluated;
3. bootstrap or hierarchical uncertainty at the animal, preparation, or specimen level;
4. end-matched, phase-randomized, amplitude-adjusted, and block-shuffled surrogates where temporal scaling is claimed;
5. ordinary morphological, graph, rate, power, and task baselines;
6. a preregistered held-out endpoint and complexity penalty; and
7. the exact source, preparation, and date class for every external parameter.

The phrase “fractal” is therefore reserved for a finite-range model that survives these controls. A positive Hurst exponent, multifractal spectrum, Rent exponent, or dendritic box-counting slope may describe one measured property without proving a common cross-scale law. The cross-scale SAN proposal is favored only when the typed feature family improves held-out reinstatement or receiver-relative prediction beyond matched ordinary features, and when the improvement disappears under the corresponding scale or morphology perturbation.

The source genealogy remains separate from this estimator contract. The 2011-2022 SAN records establish the historical problem and proposed vocabulary; the present equations, finite-range criteria, nulls, and application tests are a 2026 formalization and are not backdated.

6.15 What added prediction identifies

The population target of the nested held-out comparison can be stated exactly. Under squared-error loss, define the Bayes predictors

$$m_0(B) = E[Y \mid B], \quad m_1(B, F) = E[Y \mid B, F]. \quad (17)$$

Then the optimal population risk difference satisfies

$$\begin{aligned} \Delta_{\text{scale}}^* &= E[(Y - m_0(B))^2] - E[(Y - m_1(B, F))^2] \\ &= E[(m_1(B, F) - m_0(B))^2] \geq 0. \end{aligned} \quad (18)$$

The equality follows from conditional-expectation orthogonality. It gives the paper a clean predictive estimand: scaling features add population prediction exactly when conditioning on them changes the conditional mean on a set of nonzero probability. Finite estimators can still overfit, so Equation 18 is tested with specimen- or session-grouped held-out risk, not training error or fold-to-fold standard error. Neuroimaging cross-validation can have wide uncertainty even when the nominal number of within-subject samples is large (Varoquaux, 2018).

Equation 18 does **not** identify mechanism, direction, or intervention response. The same observational variables can arise from two structural models:

$$\begin{aligned}
\mathcal{C}_A : F &= U, & Y &= F + \varepsilon, \\
\mathcal{C}_B : F &= U, & Y &= U + \varepsilon, \\
P_A(F, Y) &= P_B(F, Y), \\
E_A[Y \mid do(F = f)] &= f, & E_B[Y \mid do(F = f)] &= 0.
\end{aligned} \tag{19}$$

The observational distributions are identical, but only the first model assigns a causal effect to intervening on (F). Multiple interventions, invariance across environments, or explicit causal assumptions are therefore required before predictive scale features are interpreted causally (Peters, Bühlmann, and Meinshausen, 2016).

6.16 Why a finite straight line does not identify a scaling law

Suppose a power law ($p(x)=x^{-D}$) is observed only at finitely many positive scales ($x_1, \dots, x_K$). For any nonzero constant (c), construct

$$q(x) = x^{-D} \exp \left[c \prod_{k=1}^K (\log x - \log x_k) \right]. \tag{20}$$

At every sampled scale, ($q(x_k)=p(x_k)$), yet (q) is not the same power law between or outside those points. This is a constructive nonidentifiability result, not an argument against finite-range biological scaling. It explains why declared intervals, off-grid validation, alternative families, resolution checks, and biologically motivated priors are part of the claim rather than optional decorations.

6.17 Correspondence maps and leakage

Relation preservation in Equation 10 depends on the cross-level correspondence map. If that map is chosen after evaluation outcomes are visible, the procedure can optimize the answer it later reports. The governed estimator is therefore

$$\hat{\Pi}_{\text{train}} = \arg \min_{\Pi \in \mathcal{P}} \sum_{j \in \mathcal{D}_{\text{train}}} \left\| R_{s+1}^{(j)} - \Pi R_s^{(j)} \Pi^\top \right\|_F^2, \quad \hat{\Pi}_{\text{train}} \text{ fixed on } \mathcal{D}_{\text{eval}}. \tag{21}$$

The same rule applies to preprocessing, fitted scale intervals, feature selection, surrogate choice, and hyperparameters. Samples from one neuron, animal, participant, session, or source item remain in one split unless the declared estimand specifically concerns within-unit generalization. Otherwise the model may learn identity rather than a transferable cross-scale relation.

6.18 Executable countermodel audit

The development-only application evaluates two grouped predictive conditions across 64 deterministic seeds. Each seed contains 48 synthetic specimens with eight rows per specimen; 36 complete specimens train the model and 12 complete specimens evaluate it. The baseline uses four ordinary

features. The nested model adds two typed scale features. No future-final seed or outcome exists in the application package.

Condition	Mean held-out Δ_{scale}	2.5%-97.5% empirical interval	Positive / negative seeds	Result
Planted scale increment	1.50865	1.14307 to 2.07334	64 / 0	recovered inside the declared generator
Redundant scale null	-0.00084	-0.00934 to 0.00832	28 / 36	not promoted

These are pipeline checks, not estimates of a biological effect. The seven countermodels supply the more important boundaries:

1. A non-power function matched a power law exactly at all five sampled scales while differing by as much as 1.0555% at the tested intermediate scales.
2. Two causal structures generated 4,096 observational rows with zero numerical difference but disagreed under ($\text{do}(F=2)$): expected outcomes 2 versus 0.
3. Row-wise specimen leakage produced mean-squared error 0.06419, whereas a complete-specimen split produced 6.17160-an apparent improvement factor of 96.15 that was entirely identity-bound.
4. Training-only correspondence selection recovered the planted map. In this particular 32-case evaluation, adaptive re-selection produced **zero** additional optimism; this countermodel is retained as a null result and does not by itself measure the broader leakage risk.
5. Phase randomization preserved the power spectrum to a maximum relative numerical difference of (3.68×10^{-16}) , showing that spectrum alone cannot identify temporal organization.
6. Amplitude adjustment preserved the sorted marginal distribution exactly while leaving a 0.01390 relative spectral difference, showing that one preserved statistic does not reconstruct the complete process.
7. Removing a quadratic nonstationary trend changed the estimated DFA slope from 0.65271 to 0.52099. The 0.13172 shift blocks interpretation as one preprocessing-invariant exponent.

The first application attempt failed before creating outcomes because three JSON-style Boolean literals appeared in Python source. The one-literal-class repair, failed attempt, unchanged frozen contract, deterministic replay, and twelve passing tests are all retained in the application evidence package. The future final-test seal remains `SEALED_NOT_OPENED` with zero accesses.

6.19 Experimental units, clustered uncertainty, and selection before outcomes

The biological tests require a second identifiability boundary: repeated cells, branches, trials, sensors, or stimulation sweeps do not create additional independent animals, donors, or participants. For experimental unit (u) and nested observation (r), the planning model is

$$Y_{u,r} = \mu + \tau X_u + b_u + \varepsilon_{u,r}, \quad b_u \sim (0, \sigma_b^2), \quad \varepsilon_{u,r} \sim (0, \sigma_\varepsilon^2). \quad (22)$$

If a unit contributes (m) nested observations, its mean has variance

$$\text{Var}(\bar{Y}_u) = \sigma_b^2 + \frac{\sigma_e^2}{m}, \quad \rho = \frac{\sigma_b^2}{\sigma_b^2 + \sigma_e^2}. \quad (23)$$

Equation 23 shows why increasing repeats can improve a unit-level estimate without replacing new independent units when the intraclass correlation (ρ) is positive. The frozen power screen therefore varies effect size, intraclass correlation, and attrition, but makes its decision in animals, donors, or participants rather than in nested observations. Let $\widehat{\mathcal{P}}(n, d, \rho, a)$ denote simulated power at candidate unit count (n), standardized effect (d), intraclass correlation (ρ), and attrition fraction (a). The planning candidate is

$$n^* = \min \left\{ n \in \mathcal{N} : \widehat{\mathcal{P}}(n, d_{\text{SESOI}}, 0.40, 0.15) \geq 0.90 \wedge \widehat{\alpha}_0(n, 0.40, 0.15) \leq 0.065 \right\}. \quad (24)$$

This is a deterministic Monte Carlo sensitivity screen, not a variance estimate from biology and not authorization to recruit humans or animals. Four protocol-level primary tests form one family. If their ordered unadjusted values are $(p_{\{1\}} \leq \dots \leq p_{\{4\}})$, Holm rejection requires

$$p_{(k)} \leq \frac{0.05}{4 - k + 1} \quad \text{for every preceding ordered test through } k. \quad (25)$$

Secondary families use a separately declared Benjamini-Hochberg procedure. Neither multiplicity rule may be changed after corresponding outcomes become visible.

6.20 Selective rescue and outcome-blind inclusion

A causal rescue must restore the route-specific relation lost under perturbation, not merely raise firing or improve behavior nonspecifically. With sham (S), predicted-route perturbation (P), matched rescue (R), and nonspecific excitation (N), define

$$L_{\text{pert}} = Y_S - Y_P, \quad G_{\text{selective}} = (Y_R - Y_P) - (Y_N - Y_P) = Y_R - Y_N. \quad (26)$$

The planned mechanism requires both a positive perturbation loss and a positive selective-rescue gain on the item-relevant relation endpoint, while retaining rate, arousal, locomotor, placement, and off-target readbacks. Outcome-blind exclusion is represented by a quality record ($\widehat{Q}_u^{\text{pre}}$) fixed before the endpoint is computed:

$$I_u = \mathbf{1}\{\widehat{Q}_u^{\text{pre}} \in \mathcal{Q}_{\text{admissible}}\}, \quad I_u \perp Y_u \mid \widehat{Q}_u^{\text{pre}}. \quad (27)$$

Equation 27 is an analysis contract, not a claim that every practical quality variable is statistically independent of outcome. It means that inclusion cannot be chosen by inspecting the result it would retain or remove.

The deterministic screen evaluated 456 design rows with 4,000 replicates per row. At the frozen worst-case planning values of 0.40 intraclass correlation and 15% attrition, it found the following screening candidates: 64 donors or animals for Protocol FCP-A at a standardized smallest effect of interest of 0.35; 96 participants for FCP-B at 0.30; 32 animals per arm for FCP-C at 0.75; and 96 participants for FCP-D at 0.30. Estimated planning power ranged from 0.95325 to 0.97250 and simulated null rates from 0.05050 to 0.05525. These values are conditional on the planning generator.

External variance evidence, ethics review, registration, and local feasibility work can increase the required numbers or stop a protocol entirely.

6.21 Synthetic multiscale reference system

The first broad development-only system has an eight-dimensional partially observed content state, four ordinary morphology variables, three typed scale variables, a receiver expectation, and a receiver-relative difference. Three transformations carry the state across scale-indexed levels. The declared generator is

$$h_{s+1} = \tanh(W_s h_s + C_s B + g U_s \text{vec}(F \otimes \delta) + r, R_s a_s - 0.18 [I_s h_s]_+) , \quad s = 0, 1, 2, \quad (28)$$

where (B) contains ordinary morphology, (F) contains typed scale variables, (δ) is the receiver-relative difference, (a_s) is recurrent anchor state, ($g=0.72$) in the planted generator and ($g=0$) in the nonfractal null, and (r) switches recurrence. The equation is a transparent synthetic mechanism, not a proposed literal cellular transfer function.

All five predictors share the same 53-column common design,

$$x_{\text{common}} = [1, h_0, B, h_0^{\odot 2}, \text{vec}(B \otimes h_0)] , \quad \dim(x_{\text{common}}) = 53. \quad (29)$$

Each then receives exactly 27 additional columns. The full model uses (F) and ($\text{operatorname{vec}}(F \otimes \delta)$); the nonlinear baseline uses 27 deterministic nonlinear functions of the common variables and never receives (F); the single-scale model uses one scale coordinate plus 18 matched padding features; the no-PWD model replaces (δ) with uncorrected receiver activity; and the correspondence control permutes (F) within each complete split. Thus

$$p_{\text{design}} = 53 + 27 = 80, \quad p_{\text{fit}} = 80 \times 8 = 640 \quad (30)$$

for every model. Equal column count is a bounded capacity control; it does not make the five inductive biases or computational costs identical. Each model is fit on 160 complete specimens and evaluated on 80 different complete specimens by fixed ridge regression,

$$\hat{\mathbf{B}}_m = (X_m^\top X_m + \lambda P)^{-1} X_m^\top Y, \quad \lambda = 0.01, \quad (31)$$

where the intercept entry of (P) is zero and all other diagonal entries are one. The preregistered primary seed-level contrast is

$$\Delta_j^{\text{primary}} = \text{MSE}_{j,\text{baseline}} - \text{MSE}_{j,\text{full}}, \quad (32)$$

and the development gate requires the lower bound of a 5,000-replicate paired seed bootstrap interval to exceed zero. For seed differences ($\Delta_1, \dots, \Delta_{40}$), bootstrap replicate (b) is

$$\bar{\Delta}^{*(b)} = \frac{1}{40} \sum_{j=1}^{40} \Delta_{J_j^{(b)}}, \quad J_j^{(b)} \stackrel{iid}{\sim} \text{Uniform}\{1, \dots, 40\}. \quad (33)$$

The interval is a deterministic analysis of declared synthetic development seeds. It is not a population interval for neurons, animals, or people.

6.22 Feature-level development results

The application produced 400 model-run rows and 40 generator-intervention rows. All fitted models used 80 columns and 640 coefficients. Positive mean squared-error differences favor the full typed-scale/PWD model.

Paired seed contrast	Mean difference	95% paired bootstrap interval	Positive / negative seeds	Development interpretation
planted baseline minus full	0.034119	0.031943 to 0.036321	40 / 0	primary planted effect recovered
planted single-scale minus full	0.023134	0.021540 to 0.024723	40 / 0	all three typed scale variables mattered in this generator
planted no-PWD minus full	0.003269	0.003046 to 0.003498	40 / 0	receiver-relative correction added bounded value
planted shuffled correspondence minus full	0.037572	0.035328 to 0.039855	40 / 0	the correct cross-level correspondence mattered
null baseline minus full	-0.000166	-0.000197 to -0.000134	3 / 37	full model was slightly worse; null not promoted

In the planted generator, the full model also improved relation-preservation correlation by 0.327109 and sign-defined content accuracy by 0.204688 over the capacity-matched baseline. These differences were positive in all 40 development seeds. The nonfractal result is equally important: when the generator removed the typed scale contribution, the full model did not manufacture an advantage and instead incurred a small held-out penalty.

The generator-level perturbation removed the typed scale route and produced mean cosine loss 0.213970 relative to sham. Exact matched restoration returned cosine similarity to 1.0 by construction; an energy-matched nonspecific injection reached 0.655607, giving a selective-rescue difference of 0.344393. Turning off recurrence changed output by mean Euclidean distance 0.016141. These are intervention and plumbing checks inside Equation 28. The exact rescue cannot be interpreted as a learned rescue policy, biological efficacy, or evidence about consciousness.

6.23 Explicit graph-temporal stress test

The next application replaced typed feature columns with explicit weighted graphs, two-dimensional node geometry, 64-step node signals, a tonic expectation, and receiver-relative departures. Thirty seeds were evaluated in a route-planted family, a geometric/spectral null, and a sign-unstable mixed-route family. Eight predictors used the same 96 design columns and 576 fitted coefficients.

In the planted family, the full graph-temporal/PWD representation beat the strong topology-and-spectrum baseline in all thirty seeds by mean control-minus-full MSE 0.12589 [0.10436, 0.14995]. It also beat graph-only, temporal-only, raw-activity/no-PWD, topology-preserving

geometry-shuffle, spectrum-preserving phase-surrogate, and nonlinear random-feature controls. That result reversed appropriately outside the planted family. The strong baseline beat the full model in all thirty geometric/spectral-null seeds by mean 1.20122, and in all thirty mixed-route seeds by mean 0.56167.

Learned compensation after removal of the receiver-relative route failed. If

$$R_s = \text{MSE}_{\text{unadapted},s} - \text{MSE}_{\text{adapted},s}, \quad (34)$$

then mean $R_s = -0.02164$, with a 95% interval of $[-0.02997, -0.01377]$, with 26 adverse and four favorable seeds. Seven of eight frozen gates passed, but the joint rule failed. Retraining a substitute representation did not restore the removed relation.

6.24 Independent-generator reversal and recovery-metric failure

A separate implementation then evaluated twenty seeds, three graph/time scales, four generator families, and seven 96-column, 576-coefficient predictors. Equal scalar counts did not equalize usable capacity. For standardized design matrix X and ridge penalty λ , effective degrees of freedom were reported as

$$\text{df}_{\text{eff}}(X, \lambda) = \text{tr} \left[X(X^T X + \lambda I)^{-1} X^T \right]. \quad (35)$$

Across the frozen systems, effective degrees of freedom ranged from 11.78 to 65.61. The strongest spectrum-topology model beat the full graph-temporal/PWD representation in all twenty hierarchical seeds: mean control-minus-full MSE -0.003558 $[-0.003912, -0.003192]$. Graph-reservoir, no-PWD, and nonlinear random-feature alternatives also won. The full model narrowly beat only the combined fixed reservoir, 0.000045 $[0.000005, 0.000092]$; the sequence-reservoir comparison remained unresolved.

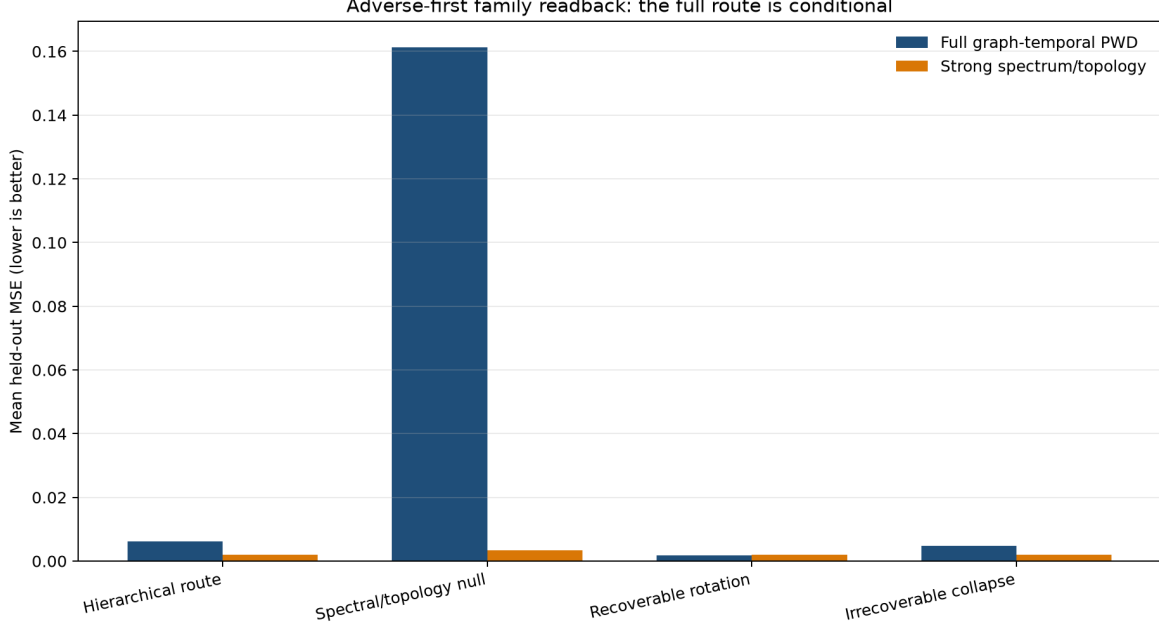


Figure 2: Adverse-first family comparison. The full graph-temporal representation is conditional rather than universally superior: the stronger spectrum-topology alternative dominates under the spectral/topology null, while other families expose smaller or reversed differences.

The same study exposed a nonidentifying recovery fraction. With intact error E_I , perturbed error E_P , and refitted perturbed error E_R , the registered statistic was

$$\rho = \frac{E_P - E_R}{E_P - E_I}. \quad (36)$$

An invertible rotation yielded exactly $\rho = 1$. Yet an information-destroying collapse produced mean $\rho = 1.205$, with a 95% interval of $[1.040, 1.437]$. The rotation-minus-collapse contrast was -0.205 , with a 95% interval of $[-0.439, -0.037]$. The fraction rewarded better post-refit prediction even when the declared route had been removed; it therefore could not identify recovery of lost information.

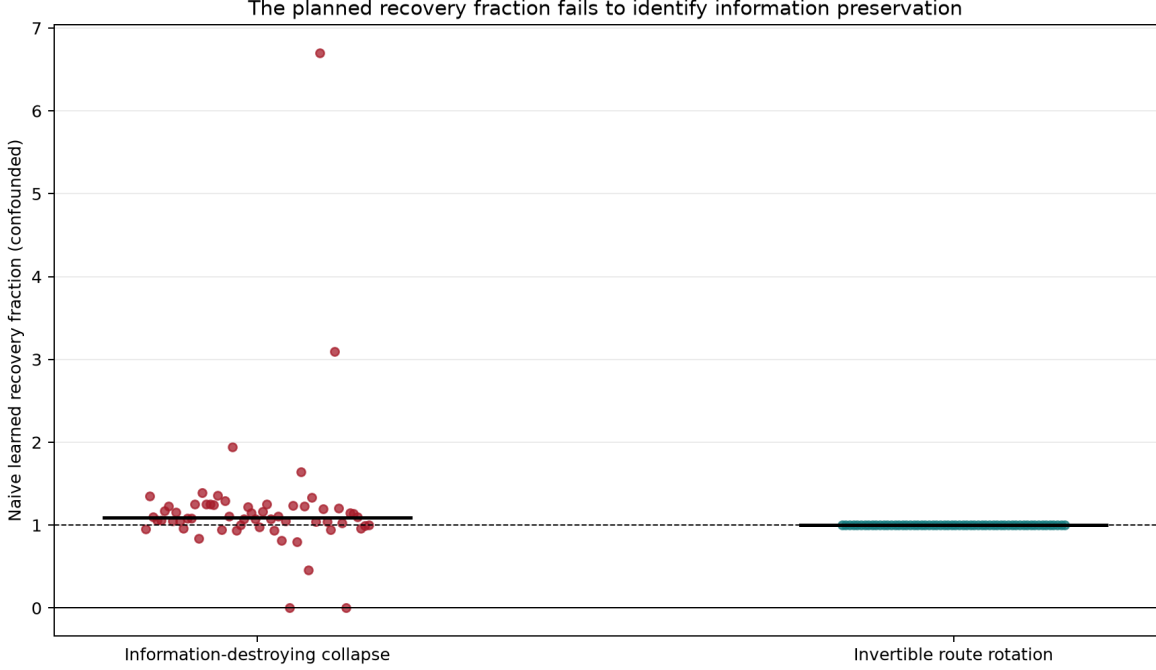


Figure 3: Recovery-metric failure. The earlier fraction returned values near or above one for both invertible transformations and information-destroying collapse, so prediction improvement alone did not identify preservation of the missing relation.

6.25 Denominator-safe formal invariants

The algebraic failure was checked in Lean without promoting the quotient itself as a recovery theorem. Define numerator and denominator separately:

$$N(E_P, E_R) = E_P - E_R, \quad D(E_P, E_I) = E_P - E_I. \quad (37)$$

The checked identity is

$$N(E_P, E_R) - D(E_P, E_I) = E_I - E_R. \quad (38)$$

Thus, whenever $E_R < E_I$, the numerator exceeds the denominator; if the denominator is positive, Equation 36 exceeds one. The formal kernel also checks squared-norm preservation and injectivity for an exact quarter-turn coordinate change, common-component preservation and noninjectivity for route erasure, and the impossibility of an exact left inverse after a nonzero route component has been erased.

Seventeen named theorems compiled under Lean 4.30.0. Six were axiom-free; eleven used only the standard kernel dependencies `propext` and/or `Quot.sound`; none used `sorryAx`, `Classical.choice`, or a user-introduced axiom. These facts explain the metric failure. They do not establish a neural variable, biological route, conscious state, or empirical mechanism.

6.26 Frozen route-identification application

The replacement endpoint measures target information unavailable from preserved common features. For route-specific target Z , common context C , and candidate route representation R , define

$$\Delta_Z = \text{MSE}(Z \mid C) - \text{MSE}(Z \mid C, R). \quad (39)$$

The new application froze four generator families, six model families, five interventions, twenty complete seeds, 4,000 seed-cluster bootstrap resamples per interval, and ten jointly required gates before any result file existed. It generated 480 model rows and 100 intervention rows in each of two complete executions. All five science outputs reproduced byte-for-byte.

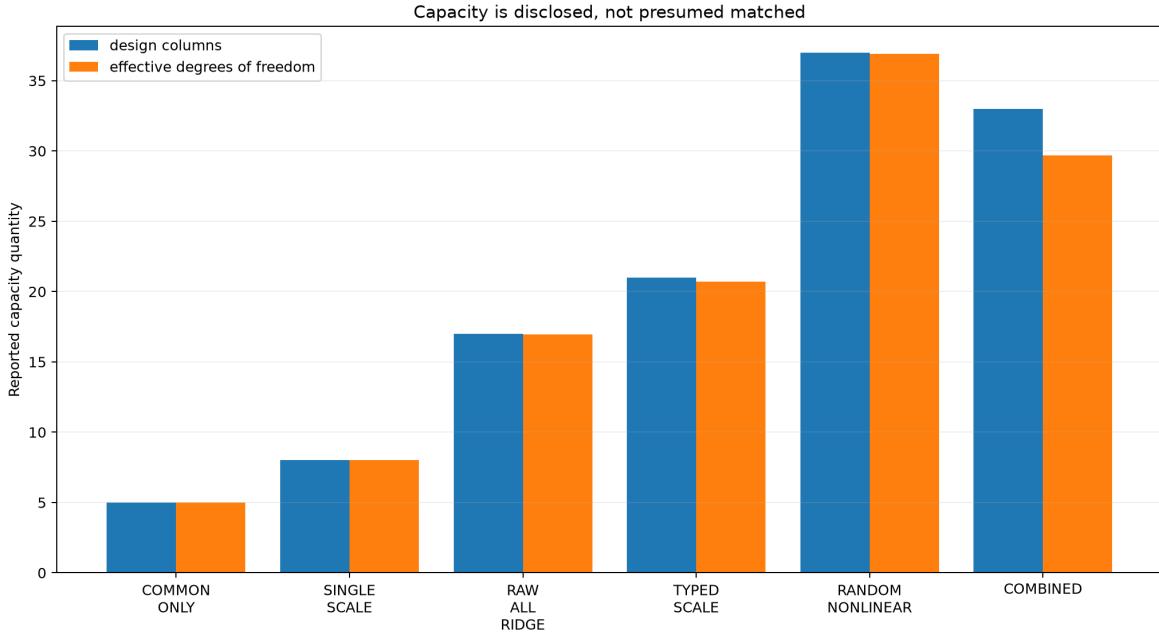


Figure 4: Capacity disclosure for the frozen route-identification application. Design-column counts and effective degrees of freedom are both reported because equal scalar counts do not imply equal usable capacity or inductive bias.

In the structured family, the typed route produced mean $\Delta_Z = 1.010752$, with a 95% interval of $[0.996080, 1.026233]$. Raw all-route ridge was slightly better on MSE, giving raw-minus-typed MSE -0.004076 , with a 95% interval of $[-0.004392, -0.003745]$, but remained inside the frozen -0.02 noninferiority margin. An invertible coordinate rotation preserved Δ_Z and route energy to floating-point tolerance. Route erasure reduced Δ_Z to zero. Exact matched restoration returned the intact value. Nonspecific restoration preserved the marginal route distribution but broke sample identity and produced $\Delta_Z = -0.039606$, with a 95% interval of $[-0.048917, -0.030889]$.

The negative controls constrained the interpretation. The typed route was adverse in the no-route family, -0.037626 $[-0.044239, -0.030978]$, and strongly adverse under decoy shift, -1.051116 $[-1.101337, -1.002614]$. When only one scale carried the target, the direct single-scale model achieved mean MSE 0.103534 , while the typed multiscale model achieved 0.795700 . The simpler correct explanation won decisively.

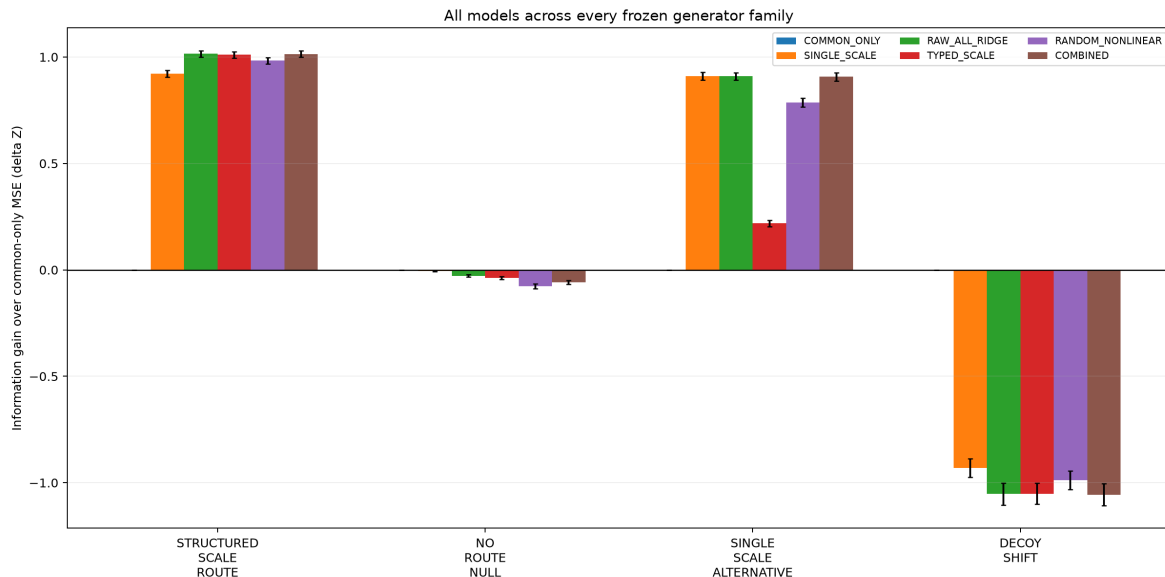


Figure 5: All six models across all four frozen generator families. A valid route is informative in the structured family, while no-route, decoy-shift, and single-scale families force refusal or selection of the simpler correct explanation.

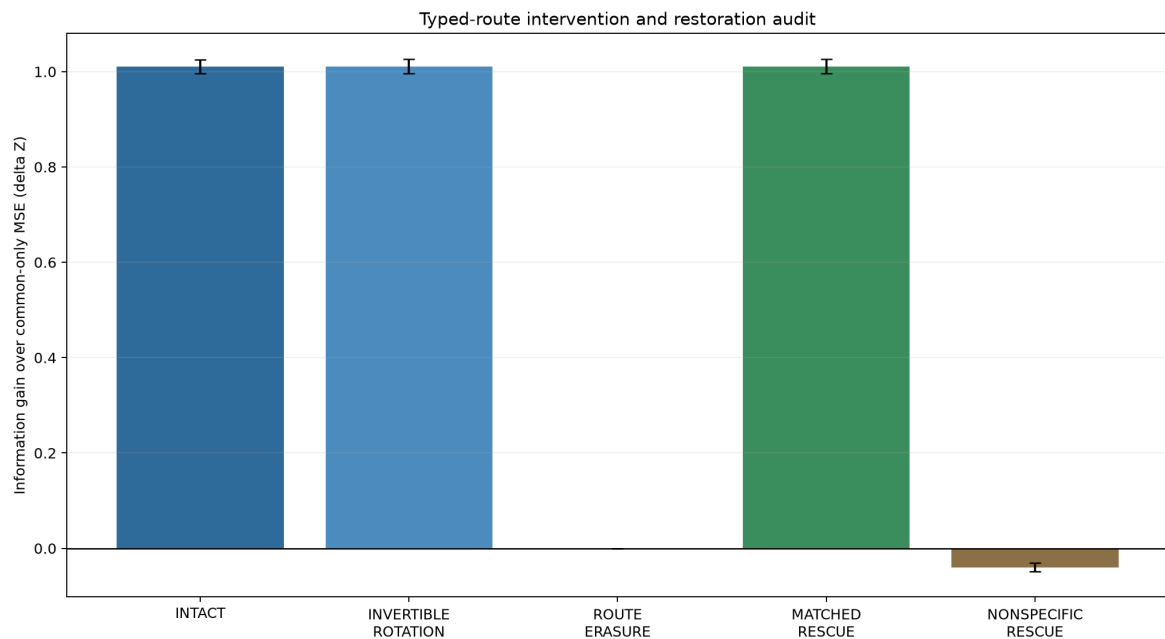


Figure 6: Route intervention audit. Invertible rotation preserves the route-specific increment, erasure removes it, exact matched restoration returns it by construction, and nonspecific sample-misaligned restoration fails.

All ten frozen gates passed jointly. The exact matched restoration and exact erasure are structural plumbing tests by construction, not learned biological repair. The result establishes bounded route

identification, transformation, erasure, restoration, and refusal behavior inside the declared synthetic generator. It does not establish universal typed-model superiority, biological FRACTAL Conscious Perception, conscious content, or therapy.

7. Testable hypotheses

H1 - Finite-range morphology

Neuronal and glial arbors will show reproducible finite-range scaling statistics after resolution, segmentation, orientation, and finite-size controls. The fitted range and estimator may differ by cell type and region.

H2 - Morphology-to-readout increment

Typed morphology scaling features will improve held-out prediction of local dendritic or somatic response beyond length, branch count, synaptic density, path distance, cell type, and recent voltage history.

H3 - Cross-scale reinstatement increment

Scaling features measured at one or more levels will improve prediction of item-specific distributed reinstatement beyond topology-only, oscillation-only, and history-aware baselines.

H4 - Asymmetric relation preservation

Successful recall will preserve selected relational structure across levels while changing carrier identity, timing, dimensionality, and spatial extent. Exact pattern identity is not required.

H5 - Content beyond global state

Typed scaling features will carry information about reinstated content after global arousal, anesthesia state, broadband power, and generic complexity are controlled.

H6 - Selective perturbation and rescue

Perturbing a route predicted to carry a cross-scale transform will selectively reduce the corresponding relation-preservation and reinstatement scores. Matched reintroduction of the missing relation, timing, or downstream state will restore them more effectively than nonspecific excitation.

8. Experimental program

8.1 Protocol FCP-A: morphology, physiology, and local readout

Use a preparation in which the same identified neurons receive three-dimensional morphological reconstruction and dendritic or somatic electrophysiology. Candidate platforms include acute slices with all-optical voltage imaging, patch-clamp recordings, and post hoc morphology. For each cell:

1. reconstruct apical and basal arbors in three dimensions;
2. quantify ordinary morphology plus $D(\epsilon)$, lacunarity, and local scaling range;
3. deliver spatially and temporally controlled synaptic or optogenetic patterns;
4. predict dendritic spikes, plateau potentials, somatic latency, waveform, and output sequence;
5. compare baseline models with and without scaling features;
6. repeat under branch-local channel or inhibition perturbation.

The independent experimental unit is the animal or human donor, never the branch, cell, stimulation pattern, or sweep. All observations from a donor remain together in the outer split, and uncertainty is donor-clustered. The frozen primary endpoint is complete-donor held-out baseline mean-squared error minus scaling-aware mean-squared error. The critical comparison is not high versus low fractal dimension by itself. It is whether the scaling vector improves prediction after ordinary morphology and conductance history are known. The predeclared causal extension uses a branch-local conductance or inhibition perturbation, route- and timing-matched restoration, and a global-excitability control.

8.2 Protocol FCP-B: distributed memory reinstatement

Use human intracranial EEG or high-density MEG/EEG during encoding and cued retrieval. Estimate item-specific reinstatement with strictly cross-validated mappings. Compute temporal scaling features from raw and amplitude-standardized signals, then evaluate:

- whether F_j predicts trial-level $\mathcal{R}_j$;
- whether it predicts correct identity and confidence after generic state features are included;
- whether the fitted scale range generalizes across sessions;
- whether the same estimator transfers across frequency bands or requires modality-specific parameters;
- whether retrieval acceleration is related to a change in transformation depth rather than simple replay speed.

The participant is the independent experimental unit. Every session, item, trial, sensor, electrode, and source estimate from that participant remains inside one outer split. MEG or high-density EEG is the confirmatory route; intracranial EEG is an opportunistic replication with a separate sampling frame and is not pooled by default. The primary endpoint is participant-level held-out item-specific reinstatement risk reduction. Phase-randomized, amplitude-adjusted, trial-shuffled, and end-matched surrogates are mandatory. Trigger failure, acquisition failure, missing identity labels, or withdrawal may exclude a unit only under the outcome-blind rule fixed before endpoint inspection.

8.3 Protocol FCP-C: multiregion engram perturbation and rescue

In a mouse memory paradigm with activity-dependent tagging, measure a local cell ensemble, downstream populations, and behavior. Use the Roy et al. distributed-engram design as a precedent, then add scale-typed features. Perturb one predicted bridge while monitoring:

- reactivation of downstream ensembles;
- relation preservation between encoding and recall;
- timing and phase residuals;
- recall behavior;
- nonspecific locomotor, fear, arousal, and firing-rate effects.

The animal is the independent experimental unit. The frozen four arms are sham, predicted-route perturbation, matched rescue, and nonspecific excitation. The primary family jointly evaluates sham-minus-perturbation loss and matched-rescue-minus-perturbation recovery of item-relevant relation preservation; Equation 26 then asks whether matched rescue exceeds nonspecific excitation. A rescue should reinstate the missing route-specific timing or downstream relational state. Global stimulation is an insufficient rescue because it cannot distinguish pathway information from excitability. Launch requires institutional animal-welfare approval, ARRIVE 2.0 planning and reporting, a harm-benefit review, refinement, humane stopping rules, and placement or expression quality criteria fixed before unblinding.

8.4 Protocol FCP-D: content-sensitive conscious access

Use a masked-visibility perceptual task while preserving forced-choice measures and, where feasible, no-report-compatible readouts. Fit separate models for global state and item content. The key test is whether scale features predict content-sensitive outcomes conditional on state, power, response bias, and perturbational complexity. The participant is the independent experimental unit, and all sessions, items, trials, visibility conditions, and sensors remain inside the participant’s outer split.

This frozen protocol does not administer sedatives or psychedelics. Existing anesthesia, sleep, or psychedelic datasets may be analyzed only as separate secondary-data studies under their original consent, acquisition, and design limits. They are not silently pooled with the masked-visibility experiment.

This experiment connects the framework to consciousness without treating any one fractal statistic as sufficient for experience.

9. Executable reference applications

The paper contains four distinct development applications, each answering a different question and retaining the result it actually produced.

1. **Feature-level multiscale system.** Forty planted-generator seeds favored the full typed-scale/PWD representation, while the nonfractal null did not. This established pipeline identifiability inside a known feature generator.
2. **Explicit graph-temporal system.** Thirty-seed graph, geometry, tonic, phase, and receiver-relative objects produced a conditional win in the route-planted family and decisive losses in

null and sign-unstable families. Learned compensation failed.

3. **Independent-generator robustness system.** A separate representation reversed the principal ranking, exposed a 5.6-fold range in effective degrees of freedom despite equal scalar counts, and showed that the earlier recovery fraction could reward route destruction.
4. **Route-identification system.** A new pre-outcome contract required route-specific information beyond preserved common features, invertible transformation recovery, erasure collapse, exact matched restoration, nonspecific-restoration refusal, null refusal, decoy-shift refusal, and recognition of a simpler single-scale explanation. All ten joint gates passed in two byte-identical executions.

Together, these applications support a conditional engineering conclusion. Typed routes can be made explicit, transformed, destroyed, restored, and refused under known generators. They do not establish that biological morphology or oscillatory dynamics instantiate the same route, and they do not establish phenomenal consciousness. The adverse studies are not discarded because the last application passed; they specify the model-capacity, generator-sensitivity, and recovery-identifiability failures that future biological work must avoid.

The executable evidence, contracts, raw rows, figures, tests, compiler output, validation receipts, and package hashes accompany this preprint. The biological protocol remains a plan. Its future evaluation seal is unopened.

10. Baselines and ablations

Every result should be compared against:

1. **size-only:** length, area, volume, branch count, and density;
2. **topology-only:** degree, modularity, path length, rich-club, and Rent exponent without morphology scaling;
3. **history-only:** recent activity and state variables;
4. **oscillation-only:** band power, phase-locking, spectral slope, and cross-frequency coupling;
5. **geometry-only:** Sholl, path, angle, and taper features without fractal estimates;
6. **single-scale:** features at the best one scale with no cross-scale operator;
7. **nonfractal matched:** same size, degree, density, spectrum, and marginal distribution with the scaling relation disrupted;
8. **topology-preserving shuffle:** geometry randomized while the graph is preserved;
9. **spectrum-preserving surrogate:** phase randomized or amplitude-adjusted signals;
10. **label and trial shuffle:** destroys content mapping while preserving global state;
11. **no-PWD:** removes receiver-relative residuals from Equation 9;
12. **no-recurrence:** replaces the telescoping sequence with a single feedforward transform.

A claim of predictive superiority requires beating the strongest capacity-aware baseline, not merely a deliberately weak comparator. A narrower route-identification claim may survive without universal superiority only when the route contributes information beyond preserved context, responds selectively to transformation and destruction, is restored by the declared matched intervention, and is refused under null, decoy, nonspecific, and simpler-alternative controls.

11. Failure conditions

The framework makes itself vulnerable to informative failure.

1. The candidate scaling range disappears after resolution, segmentation, threshold, finite-size, or orientation controls.
2. A lognormal, exponential, stretched-exponential, or ordinary geometric model fits as well as or better than the proposed scaling law.
3. The estimated exponent or spectrum changes materially with reasonable preprocessing choices.
4. Fractal and multifractal features add no held-out prediction beyond variance, power, morphology, topology, and history.
5. Different modalities require incompatible descriptions and no typed cross-scale mapping improves prediction.
6. Relation preservation does not distinguish correct from incorrect retrieval.
7. A perturbation changes firing rate or arousal but not the route-specific relation predicted by the model.
8. Rescue succeeds equally well through nonspecific excitation.
9. Scale features distinguish global state but not perceptual or memory content.
10. The new model adds no value beyond the existing Telescoping Network Reinstatement, Morphology-to-Phase, and Cross-Scale Approximation models.

Any such result narrows the scale, preparation, estimator, or functional role that the framework should claim. The claim ledger records the exact boundary and retains the source genealogy.

12. Nonduplication and relationship to neighboring SAN papers

This paper has one scientific center: **the added predictive and causal value of measured scaling variables for cross-scale reinstatement**.

Telescoping Network Reinstatement in Self-Aware Networks owns the broader local-to-population reconstruction chain and a finite reconstruction application. *From Morphology to Phase* owns morphology-conditioned local recognition and PWD readout. *From Neural Trajectories to Cross-Scale Approximation* owns typed mathematical transport across levels. *From Cellular Difference to Population Reinstatement* owns projection, inhibition, amplification, and population-scale re-expression. *The Qualia Tomogram* owns the general receiver-conditioned forward and inverse reconstruction problem.

The present paper contributes something those works do not: estimator-specific fractal, lacunarity, multifractal, long-range-dependence, and Rentian variables; finite-range and surrogate requirements; nested held-out comparison against ordinary morphology and topology; and a route-specific perturbation/restoration test of whether typed finite-range relations explain functional reinstatement.

If the paper reduces to generic projection, reconstruction, PWD, recurrence, or tomography without independently measuring scale structure and route-specific added value, it should be absorbed into the neighboring paper that owns that operation. Its standalone burden is the conjunction of typed finite-range scaling, incremental information beyond preserved context, destructive intervention, and selective route restoration.

13. Apex semantic alignment with existing science

The highest-fidelity shared representation is straightforward. Existing neuroscience measures branching morphology, nonlinear dendritic integration, network topology, long-range temporal dependence, distributed engrams, neural reinstatement, and state-dependent complexity in different preparations. SAN joins those established objects into a proposed operation: learned local structure conditions receiver-relative differences that are transformed across recurrent arrays into a distributed reconstruction.

The external literature need not use SAN vocabulary to measure components of that chain, and the SAN source need not predict every later estimator or molecular result to retain its authored mechanism. The alignment preserves both layers:

- established studies define bounded morphology, topology, dynamics, reinstatement, and consciousness-state results;
- SAN supplies the staged receive-transform-project and synapse-to-network reconstruction hypothesis;
- the present paper adds a preregisterable test of scaling variables' incremental and causal value.

Where no experiment yet joins those layers, the status is **APEX-ALIGNMENT-OPEN**. Source, variable, scale, and mechanism translation continues, discriminating experiments are specified, and the program proceeds on the premise that a higher-fidelity alignment exists and must be recovered.

14. Discussion

The strongest reading of FRACTAL Conscious Perception is neither mystical nor cosmetic. It is an engineering claim about how relations can survive changes of carrier and scale. A dendrite, a cortical column, and a whole-brain pattern need not resemble one another visually. They may still implement a sequence of transforms in which learned local structure changes the probability, timing, and content of wider recruitment.

This perspective also explains why exact self-similarity is the wrong target. Biological systems have boundaries, characteristic lengths, heterogeneous cell types, and finite sampling. A dendritic arbor may exhibit useful effective scaling over less than two orders of magnitude. A connectome may show Rentian scaling over a partition range. An EEG amplitude envelope may show long-range dependence over selected windows. These measurements can coexist without a universal exponent because they characterize different objects.

The functional requirement is more demanding than any of them alone. A scaling feature must predict something the ordinary variables did not already predict. It must generalize to held-out data, survive nulls, and ideally support a selective perturbation and rescue. That criterion protects the theory from visual analogy while preserving the original intuition that memory moves from microscopic learned structure into distributed conscious activity through recursive transformations.

The framework also clarifies the role of oscillations. Oscillations provide repeatable temporal relations and opportunity windows, while receiver-relative departures can alter which path is recruited. Scaling statistics describe organization across windows or levels; they do not replace the specific content carried by phase, amplitude, timing, cell identity, morphology, and synaptic history. Conscious content arises in the differentiated reconstruction, not in an exponent considered alone.

Finally, the paper makes “telescoping” operational. A telescope changes scale while preserving relations needed to identify its target. Neural telescoping is asymmetric: each receiving population compresses some distinctions, amplifies others, adds contextual predictions, and routes action-relevant consequences. The empirical question is whether finite-range scaling helps that operation and, if so, where.

The application sequence changes what a responsible answer looks like. One generator rewarded the full representation, another reversed the ranking, and a third showed that raw ridge could decode the structured target slightly better than typed alignment. This is not a contradiction among results; it is evidence that generator, representation, capacity, and endpoint jointly determine what can be identified. Typed coordinates add scientific value when they make a proposed route inspectable and support selective interventions, even when they are not the lowest-error universal predictor.

The single-scale alternative is especially important. The multiscale model should not win merely because the paper is about multiscale organization. When one scale alone carried the target, the direct single-scale model won decisively. Likewise, the no-route and decoy-shift families made the typed representation worse than preserved common context. Those refusals show that the endpoint is sensitive to the presence, absence, and stability of the declared relation.

The matched-restoration result remains deliberately modest. Restoring the exact sample-aligned route returns the intact value because the route is restored by construction. Its scientific role is to validate identity-sensitive measurement and to contrast matched restoration with a marginally identical but sample-misaligned route. Demonstrating biological recovery would require a system that learns or regenerates an effective route after damage, along with evidence that the route-specific information-not merely aggregate task fit-has returned.

15. Limitations and next gates

The complete program remains bounded by its data. No application in this paper measures a living dendritic arbor, biological scale-bridging route, conscious percept, or clinical outcome. The synthetic generators are transparent tests of estimators and causal logic, not replicas of neural tissue. Their favorable results identify what the code can recover when its assumptions hold. Their adverse results identify where those assumptions fail.

The main unresolved limitations are:

- joint datasets containing cellular morphology, physiology, network activity, intervention, behavior, and content-sensitive report remain uncommon;
- finite-range estimators are sensitive to resolution, segmentation, preprocessing, sample size, and the selected scale window;
- morphology, graph topology, temporal dependence, oscillatory coordination, and functional reinstatement are different objects and may not share one exponent or even one useful family of estimators;
- correspondence maps between cells, populations, signals, and behavior may be only partially identifiable;
- intervention tools can change excitability, arousal, metabolism, movement, or measurement quality beyond the proposed route;
- conscious report and memory retrieval introduce selection, response, and task confounds;

- equal columns or parameters do not equalize effective capacity, inductive bias, optimization, compute, or memory;
- the latest matched restoration is exact by construction and does not demonstrate learned recovery;
- raw ridge and combined features slightly outperform typed alignment in the structured route generator;
- the strongest independent-generator study remains adverse to the full graph-temporal/PWD representation;
- a scale-specific contribution may be modality-, preparation-, task-, or state-specific rather than universal.

The next empirical gate is not another synthetic retuning cycle. It is an independently specified measured system in which a route-specific variable is labeled without using the candidate model, preserved common features cannot reconstruct it alone, strong architecture-native alternatives are retained, and intervention/restoration acts on the proposed route. Complete animals, participants, preparations, or recording sessions-not cells, windows, trials, or augmented fragments-must define the uncertainty unit unless the design justifies another independent unit.

No confirmatory protocol can launch from this manuscript alone. A future site must replace planning variances with external or pilot evidence, complete feasibility and safety review, obtain all required human or animal ethics approvals, precommit exclusions and multiplicity, secure evaluation custody, and demonstrate that the measurement and intervention can satisfy the refusal conditions. The biological final-test seal in this package remains `SEALED_NOT_OPENED`.

16. Conclusion

A microscopic memory trace does not have to become a literal microscopic movie. It has to change what receiving cells and populations do. Those receivers transform the signal according to their learned structure and current state; recurrent exchange preserves some relations, suppresses others, and makes a distributed pattern available for recognition, report, or action. FRACTAL Conscious Perception names the hypothesis that measured finite-range organization contributes predictive and causal value to that asymmetric transformation.

The hypothesis is now precise enough to discriminate among explanations. Morphology, topology, temporal dynamics, and reinstatement are measured separately. Finite-range fits require alternative distributions and surrogates. Scaling features must add held-out information beyond preserved common features and strong alternatives. Coordinate changes must be separated from route destruction. Restoration must return the missing relation rather than merely improve aggregate prediction.

The synthetic record is intentionally mixed. It contains planted wins, null and misspecification losses, failed learned compensation, an independent-generator reversal, a formally verified metric failure, and a new route-identification system that passes all ten frozen transformation, erasure, restoration, and refusal gates. That last result supplies a working reference architecture, not biological proof. The next decisive step is measured route-specific evidence under independent custody. The scientific contribution of the present paper is to make that step explicit, executable, and difficult to counterfeit.

17. Data, code, and reproducibility

The companion package includes the source genealogy and primary-source ledger; claim-source matrices; all synthetic generators, frozen contracts, unit tests, raw result tables, model and intervention summaries, figures, and validation receipts; the Lean source, compiler output, axiom audit, and claim signature; and SHA-256 manifests for every promoted artifact. The route-identification study was executed twice from its frozen contract, and all five science outputs reproduced byte-for-byte.

The future biological protocol, power plan, and seal are included for transparency. No biological payload or result was accessed. The synthetic data are generated by the disclosed programs and should not be represented as neural recordings.

18. Ethics and claim boundary

This paper reports no human-participant experiment, animal experiment, clinical intervention, or therapeutic recommendation. Any future biological implementation requires independent scientific review, appropriate ethics approval, feasibility and safety work, and a new preregistered custody chain. Formal verification establishes only the declared mathematical invariants. Synthetic application success establishes only behavior within the declared generators.

The paper therefore distinguishes four claim classes: dated SAN genealogy; external scientific evidence for constituent measurements; mathematical and software results within declared models; and future biological hypotheses. Evidence does not transfer automatically from one class to another.

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